Literature connections: word statistics, charge, and L-positivity for cylindric partitions

Notes for research

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Abstract

We collect four threads from the literature that connect to the L-positivity of the cylindric partition frequency generating function $\widetilde{\mathrm{GK}}_c \in \mathrm{QSym}$ established in our companion note (QUASI_SYMMETRIC.tex). The goal is to identify the right combinatorial framework for understanding this positivity and its relationship to Warnaar's Conjecture 2.7.

1 The charge statistic (Lascoux–Schützenberger)

Lascoux and Schützenberger [**?**, **?**] introduced the *charge* statistic on words and semistandard Young tableaux. For a tableau $T \in \mathrm{SSYT}(\lambda, \mu)$ (shape λ, content μ), the charge $\mathrm{charge}(T) \in \mathbb{Z}_{\geq 0}$ is computed from the *reading word* $w(T)$ as follows:

1. Read the entries of T from right to left, bottom to top, to obtain the word $w(T) = w_1 w_2 \cdots w_n$.
2. Decompose $w(T)$ into "standard subwords" by successively extracting the leftmost occurrence of $1, 2, \ldots, \mu_1$, then repeating.
3. The charge of each standard subword is $\sum_{i \in \mathrm{Des}(\sigma)} (n - i)$, where σ is the subword read as a permutation and Des denotes its descent set.
4. $\mathrm{charge}(T)$ is the sum of the charges of all subwords.

The main theorem is that Kostka–Foulkes polynomials satisfy

$$K_{\lambda,\mu}(q) = \sum_{T \in \mathrm{SSYT}(\lambda,\mu)} q^{\mathrm{charge}(T)}.$$

Question 1. *Is there an analog of the charge statistic for cylindric partitions? Specifically, given a CPP $\pi = (\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$, can one define a reading word $w(\pi)$ and a charge-like statistic such that the $Q_{n,c}(q)$ polynomials (or the L-coefficients of $\widetilde{\mathrm{GK}}_c$) are given by a charge-weighted sum?*

Remark 2. *We have previously verified computationally that $Q_{n,c}(q)$ does* not *decompose as a nonneg sum of Kostka–Foulkes polynomials $K_{\lambda,\mu}(q)$. However, the QSym framework may require a* different *charge-like statistic adapted to compositions rather than partitions.*

2 The plactic monoid and crystal bases

The *plactic monoid* (Lascoux–Schützenberger [**?**]) is the quotient of the free monoid on positive integers by the Knuth relations:

$$\begin{aligned} xzy &\equiv zxy \quad \text{if } x \le y < z, \\ yxz &\equiv yzx \quad \text{if } x < y \le z. \end{aligned}$$

Two words are *plactically equivalent* if and only if they have the same RSK insertion tableau. The plactic monoid provides a combinatorial model for the representation ring of GL_n:

$$s_\lambda \cdot s_\mu = \sum_\nu c^\nu_{\lambda\mu}\, s_\nu,$$

where the Littlewood–Richardson coefficient $c^\nu_{\lambda\mu}$ counts plactic classes of words with certain content and RSK shape conditions.

In the language of Kashiwara's crystal bases, the plactic monoid is the monoid of the crystal graph $B(\infty)$ for type A. The crystal operators $\tilde{e}_i, \tilde{f}_i$ act on words and preserve the RSK recording tableau while modifying the insertion tableau.

Question 3. *The descent set of a word is a plactic invariant (it depends only on the RSK insertion tableau via the descent set of the corresponding standard tableau). Since $\widetilde{\mathrm{GK}}_c$ is L-positive and L_α encodes descent sets, can the L-coefficients $b_\alpha(c)$ be interpreted as counting plactic classes of "cylindric words" with descent composition α?*

Remark 4. *For type A_2 (rank 3), the crystal graph has a natural connection to the three partitions $(\nu^{(1)}, \nu^{(2)}, \nu^{(3)})$ in a CPP via the weight lattice of $\mathfrak{sl}_3$. The cylindric condition (cyclic interlacing) may correspond to a periodicity condition on crystal paths, connecting to the* affine *crystal $\widehat{B}(\Lambda)$ for $\widehat{\mathfrak{sl}}_3$.*

3 Keys and Demazure atoms (Lascoux)

Lascoux [**?**] introduced *keys* (also called *Demazure characters* or *key polynomials*) κ_α indexed by compositions α, and their "atomic" decomposition: the *Demazure atoms* $\mathfrak{A}_\alpha$ satisfy

$$\kappa_\alpha = \sum_{\beta \leq \alpha} \mathfrak{A}_\beta,$$

where $\leq$ is the Bruhat-like order on compositions.

The Demazure atoms form a basis of the polynomial ring $\mathbb{Z}[x_1, \ldots, x_n]$ (in finitely many variables) that is closely related to the fundamental quasi-symmetric functions: under the limit $n \to \infty$, the atoms stabilize to the L_α.

More precisely, Mason [**?**] showed that the *quasi-key polynomials* (also called *Demazure atoms*) map to fundamental quasi-symmetric functions under the forgetful map from polynomials to quasi-symmetric functions.

Question 5. *Can* $\widetilde{\mathrm{GK}}_c$ *be lifted from* QSym *to the polynomial ring as a positive sum of Demazure atoms? If so, the extra structure (dependence on a specific number of variables) might encode the max-entry information that is lost in the QSym framework.*

4 P-partitions and labeled posets (Stanley, Gessel)

Stanley's theory of P-partitions [**?**] provides the foundational connection between labeled posets and quasi-symmetric functions. Gessel [**?**] showed that the generating function for P-partitions of a labeled poset (P, ω) is always a nonneg sum of fundamental quasi-symmetric functions:

$$\Gamma_{(P,\omega)}(\mathbf{z}) = \sum_{\substack{f: P \to \mathbb{Z}_{>0} \\ f \text{ is a } (P,\omega)\text{-partition}}} \prod_{p \in P} z_{f(p)} = \sum_\alpha c_\alpha L_\alpha, \qquad c_\alpha \in \mathbb{Z}_{\geq 0}.$$

A (P, ω)*-partition* is an order-preserving map $f : P \to \mathbb{Z}_{>0}$ that satisfies strict inequalities $f(p) < f(q)$ whenever $p <_P q$ and $\omega(p) > \omega(q)$ (i.e., at "descents" of the labeling).

Question 6. *Can cylindric partitions be realized as* P*-partitions for a suitable labeled poset? Concretely:*

1. *Define a poset P_c whose elements are the positions (i, j) (partition i, row j) of a CPP with profile c.*

2. *The interlacing conditions $\nu_j^{(i)} \geq \nu_{j'}^{(i')}$ define order relations on P_c.*

3. *A labeling $\omega : P_c \to \{1, 2, \ldots\}$ determines which inequalities are strict vs. weak.*

4. *A CPP is then exactly a (P_c, ω)-partition, and L-positivity of $\widetilde{\mathrm{GK}}_c$ follows from Stanley–Gessel theory.*

The key subtlety is the *cyclic* nature of the interlacing: $\nu_j^{(3)} \geq \nu_{j+c_0}^{(1)}$ closes the cycle $1 \to 2 \to 3 \to 1$. A standard (acyclic) poset cannot directly model this. However, by unrolling the cycle (working with a finite number of values), the resulting poset may be acyclic, and the P-partition framework may apply.

Remark 7. *If Question* **??** *has a positive answer, then the L-positivity of $\widetilde{\mathrm{GK}}_c$ would follow from general theory, and the L-coefficients b_α would count* linear extensions *of P_c with descent set $D(\alpha)$. This would provide a concrete combinatorial interpretation of the b_α.*

5 Summary and connections

The four threads above suggest the following picture:

Framework	**Key object**	**Connection to CPPs**
Charge (LS)	$q^{\mathrm{charge}(T)}$ on SSYT	Refines $K_{\lambda,\mu}(q)$; need cylindric analog
Plactic (LS)	Crystal graph $B(\lambda)$	Affine crystals for $\widehat{\mathfrak{sl}}_3$ model cylindric tableaux
Keys (Lascoux)	Demazure atoms $\mathfrak{A}_\alpha$	Lift from QSym to polynomials; may encode max-entry
P-partitions (SG)	Labeled poset (P, ω)	CPP interlacing $\approx$ poset; cyclic closure is the obstacle

The most promising direction for proving L-positivity appears to be the P-partition approach (Section **??**), as it would give L-positivity directly from general theory. The charge/crystal approaches may be more relevant for the original $Q_{n,c}(q)$ positivity problem, where the max-entry variable must be incorporated.

References

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